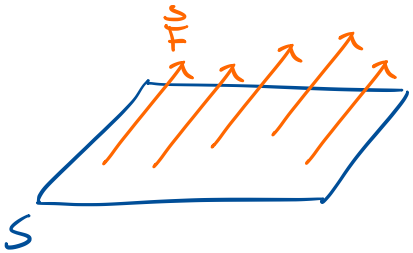
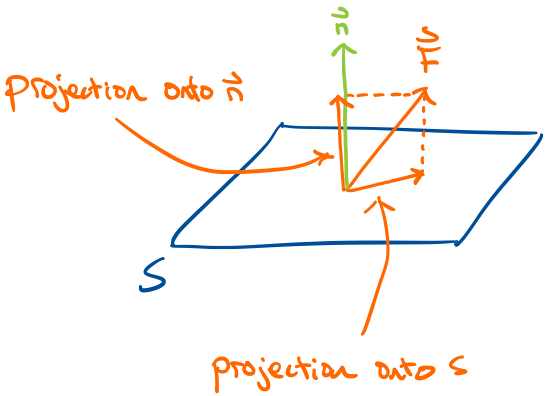
19.1 Flux integrals

If S is a surface and $\vec{F}$ is a vector field, the flux of $\vec{F}$ through S is a measure of "how much" of the field flows through S.



For simplicity, first assume S is a flat surface, $\vec{F}$ is constant, and $\vec{n}$ is a unit normal vector to S.



Then:

$$\text{flux through } S = (\vec{F}\cdot\vec{n})(\text{area of } S)$$

length of proj. of $\vec{F}$ onto $\vec{n}$

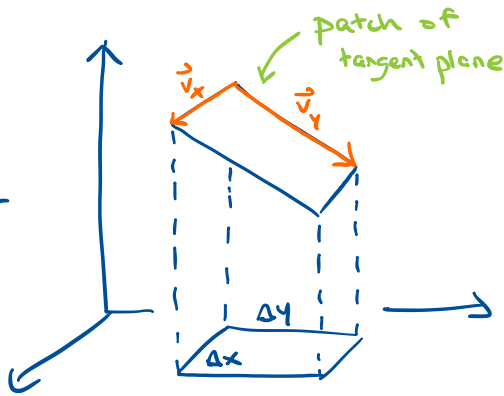
The vector $\vec{A} = (\text{area of } S)\vec{n}$ is called the area vector of S. Thus:

$$\text{flux through } S = \vec{F}\cdot\vec{A}.$$

→ Do problems 1-4.

What if S is not flat or $\vec{F}$ is not constant?

Suppose S is the graph of a function $z = f(x,y)$. We approximate S by using patches of tangent planes. The area vector of one of these patches is



$$\Delta\vec{A} = \|\vec{v}_x \times \vec{v}_y\|$$
$$= \langle -f_x, -f_y, 1\rangle \Delta x \Delta y.$$

(with upward orientation!)

$\vec{v}_x = \langle 1, 0, f_x\rangle \Delta x$
$\vec{v}_y = \langle 0, 1, f_y\rangle \Delta y$

Thus the flux is approximately

$$\sum \vec{F}\cdot\Delta\vec{A} = \sum \vec{F}(x,y,f(x,y))\cdot\langle -f_x, -f_y, 1\rangle \Delta x \Delta y.$$

Passing to the limit, we get the exact flux integral:

$$\int_S \vec{F}\cdot d\vec{A} = \int_{y_0}^{y_1}\int_{x_0}^{x_1} \vec{F}(x,y,f(x,y))\cdot\langle -f_x, -f_y, 1\rangle \, dx\, dy$$

So if S is the graph of f, then

$$d\vec{A} = \langle -f_x, -f_y, 1\rangle \, dx\, dy$$

vector area element

→ Do problems 5 and 6.